# Ali Husain

## Software Engineer and Product Designer

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## Summary

I design and build complete systems: the interface people use, the data and services behind it, and the infrastructure that keeps it running. Offline-first apps that need no account and no server, a 15-service platform that restores its own backups every week to prove they work, a workbench that judges language models blind, and guardrails for AI agents. Each one is specified first, tested before it ships, and documented to run without me; 1,520 automated tests run in CI across the public repositories. Open to remote contracts and to AI training and evaluation work (code review, model-output rating, prompt and rubric writing). UTC+3 overlaps the full European working day.

## Skills

**AI and LLM engineering:** agents and multi-agent workflows, LLM evaluation (blind pairwise judging, automatic checks, regression tracking), prompt engineering, local model deployment, retrieval and search, MCP. Anthropic and OpenAI models, open-weight LLMs, local GPU inference.

**Cloud, DevOps and SRE:** Linux (Debian, Arch), Docker, systemd, Git, GitHub Actions, CI/CD, infrastructure as code, monitoring, incident response. Data: SQLite, Redis, CouchDB.

**Networking and security:** DNS, TLS, Caddy, Nginx, mesh VPN (Tailscale, WireGuard), firewalls, intrusion prevention, encryption, secrets handling, privilege separation.

**Backup and disaster recovery:** verified backups, multi-site copies, automated restore drills, rebuild runbooks.

**Software development:** Python, TypeScript, JavaScript, Go, C, C++, Bash, SQL.

**Web and product:** Astro, React, Preact, FastAPI, Tailwind, PWAs, offline-first design, accessibility (WCAG 2.1 AA), Playwright, Vitest, Lighthouse CI, front-end and UX.

**Video, motion and audio:** editing, beat-matched cutting, color grading, motion graphics, 4K mastering; music sync, narration, sound design, mixing to loudness targets. Image, video, voice and music generation models; programmatic editing with Python and ffmpeg.

**Automation and documentation:** scheduled services, webhooks, APIs; technical writing, runbooks, data dashboards.

## Experience

### AI Systems Engineer

Independent · agents, evaluation and model deployment · github.com/gallowforge/evalbench, github.com/gallowforge/aih, github.com/gallowforge/agent-guardrails

- Built Evalbench, a self-hosted LLM evaluation workbench (Python, FastAPI, React): blind pairwise judging, seven automatic check types including code tests in a sandbox with no network, and regression tracking across local and API models; 171 automated tests.
- Designed and run an autonomous AI operations agent that monitors and maintains production infrastructure around the clock, bound by a written charter, woken by alerts, with every change logged.
- Built multi-agent delivery pipelines of up to 9 coordinated agents (1.45M tokens, 566 tool calls) with build, critique and automated browser-verification stages, and ran 13-agent system audits.
- Built a knowledge pipeline and agent guardrails: 4 GB+ of conversations from three machines in one full-text archive, LLM extraction cross-checked by a second model, and a hook that screens agent commands.

### Infrastructure and DevOps Engineer

Independent · self-hosted production platform, run as code · github.com/gallowforge/homelab, github.com/gallowforge/vesper

- Operate a 15-service containerized platform (14 health-checked) behind one TLS edge: DNS, password and identity management, file sync, smart-home control; remote access over a 7-node mesh VPN with zero port forwards.
- Built Vesper, a read-only operations dashboard: one 7.4 MB Go binary, a TypeScript UI over Server-Sent Events, a real 24-hour history of every service and honest "can't see" states, privilege-split systemd services; 191 automated tests, including a 34-check end-to-end suite under real systemd.
- Designed backup and disaster recovery: 12 integrity-verified sets nightly, 3 copies on 3 machines, automated weekly restore drills, a 2 to 3 hour bare-metal rebuild plan.
- Built observability for 77 failure conditions plus 13 uptime probes at 60 s, self-healing watchdogs and a dead-man switch. All as code: more than 400 commits, 21 scheduled services, drift detection, about 4,100 lines of runbooks and notes.

### Web and Product Developer

Independent · product design, front-end and delivery · github.com/gallowforge/Ethereal, github.com/gallowforge/ballast

- **Ethereal** (live at gallowforge.github.io/Ethereal): led product design and delivery of an account-free, offline-first health and fitness app that plans the day, coaches workouts and tracks nutrition. 41 animated exercises, an offline database of 1,474 foods and 397 automated tests, built to WCAG 2.1 AA and Lighthouse 100. TypeScript, Astro, PWA.
- **Ballast** (live at gallowforge.github.io/ballast): an offline-first personal finance app that turns savings into a monthly salary that lasts. Reads bank statements on the device and reconciles every row against the running balance. No account, no server; 599 automated tests. TypeScript, Astro, Preact, pdf.js.
- **Sites, 3D and 40+ interactive tools:** a six-page animated brand site for an AI video-ad studio (Next.js, Tailwind, Framer Motion); a scroll-driven WebGL story page verified by automated screenshots; live market dashboards, learning tools, data visualizations and planners, each designed for phone and desktop.

### Video, Audio and Generative Media Producer

Independent · production method, editing, sound and delivery · github.com/gallowforge/filmforge

- Built filmforge, a documented production method for painted films made with AI image and video models: nine Claude Code skills (story, style lock, prompt grammar, frame QA, motion, sound, delivery) and seven tested Python tools that price, lint and verify the work at every stage gate.
- Post-production built in code: beat-matched editing, color grading, motion graphics and animated titles, narration timed across cuts, layered sound design and mixing to loudness targets, 4K mastering on GPU.
- Delivery checked, not assumed: every encode verified for streams, loudness and picture-to-music sync, and each production archived with a SHA-256 completeness proof anyone can re-verify.

## Open-source projects

- **ballast** (github.com/gallowforge/ballast): offline-first personal finance app. 599 tests.
- **Ethereal** (github.com/gallowforge/Ethereal): offline-first health and fitness coach. 397 tests.
- **vesper** (github.com/gallowforge/vesper): read-only homelab operations dashboard with a 24-hour service history. 191 tests. Live demo at gallowforge.github.io/vesper.
- **evalbench** (github.com/gallowforge/evalbench): self-hosted LLM evaluation workbench. 171 tests. Live demo at gallowforge.github.io/evalbench.
- **homelab** (github.com/gallowforge/homelab): 15 self-hosted services run like production. 43 tests.
- **filmforge** (github.com/gallowforge/filmforge): a production method for AI-assisted films. 40 tests.
- **agent-guardrails** (github.com/gallowforge/agent-guardrails): guardrails for AI coding agents. 36 tests.
- **aih** (github.com/gallowforge/aih): every AI chat in one local full-text archive. 31 checks.
- **bwtidy** (github.com/gallowforge/bwtidy): audit and tidy a password vault locally. 12 tests.

## Education

**Computer Science and Chemical Engineering**, Kuwait University

## Learning

Continuous self-study in algorithms, computer science and AI, each put to use in a shipped project.